Chapter 1

Analytical and numerical modelling of wave propagation in (meta)poro(elastic) layers

Contents

This chapter lays the foundation for the thesis by providing the fundamental theoretical notions and methods used in this work, in order to guide the reader through the various chapters that will follow. At first, an overview of the acoustic modelling of porous media is proposed. Following the presentation of the state of art in the introduction and of the areas of focus that have been identified from the latter, the key notions and methods for the computation of the scattering of waves by metaporous structures will be presented. From there, the formalism for the calculation of dispersion curves will be presented along with an overview of the methods used for such computations. Both of these topics will be illustrated with preliminary results in simple configurations.

1.1 Wave propagation in homogeneous layers

In the previous section, a formulation for rigid-frame porous media and two formulations for poroelastic media modelling have been presented. Concluding the presentation of these motion equations, some characteristics of the wave propagation in unbounded porous and poroelastic media have been shown. The present section proposes the introduction of a geometry, that is to say boundaries or interface conditions delimiting different domains characterized each by a different type of material. Generally, the configurations studied in this work assume a monochromatic plane wave originated from a fluid half-space, i.e. air, impinging at the surface of a structure and is then scattered (reflected and transmitted) from and through each interface. The existence of an interface leads to the propagation of not only the bulk waves but also of so-called surface waves[1] where the interface itself acts as the support for propagation. Different types of waves propagate along the interface. An example can be surface water waves that follow the interface between air and water; all the energy is localized at the interface. A similar behaviour exist for seismic waves where the interface of a solid substrate supports the wave propagation, that is Rayleigh (longitudinal motion) and Love waves (transverse motion) which both show an energy decaying with depth. Scholte waves are waves that propagate along the interface between a fluid and a solid medium.

If the material in which the wave propagates is bounded by two parallel, free, and flat surfaces, i.e., a plate, then it is refered to as a layer. The waves supported by the medium propagate by reflecting alternatively from one surface to the other; they are said to be guided along the layer. These observations of successive reflections and refractions lead to the concept of waveguides, but their analytical descriptions often leads to writing complex system that can be difficult to solve. To determine the modes of a structure, one must seek to solve the propagation equations and the boundary conditions imposed at its boundaries. In a first section, the scattering of a plane wave by a homogeneous porous layer is investigated along with the various method available. Next, the methods for the computation of the modes of these configurations are presented. In a subsequent section, more complex geometries, those of metaporous structures, are of interest and both methods for the computation of the scattering coefficients and the modes of those configurations will be discussed.

1.1.1. Analytical plane wave modelling

Modeling acoustic propagation in homogeneous layers by assuming plane waves is the simplest formalism in acoustics. We assume these plane waves such that they satisfy the motion equations, e.g. the ones introduced in **??** or **??**. Introducing a simple problem, that of the scattering of an incident plate by a porous layer, we present an example of plane wave modelling for a simple system. The system is 2D with coordinates $\boldsymbol{x} = (x_1, x_2)$ constituted of a porous layer with thickness h and thus two interfaces of infinite extent along x_1, there is a fluid-porous interface at $x_1 = h$ and a porous-fluid interface at $x_1 = 0$. Both media of the air half-spaces and porous layer are modeled as fluid, thus the interfaces conditions should impose continuity of the pressure fields and normal particle displacements.

Assuming a $e^{\mathrm{i}\omega t}$ time-dependance, the incident pressure field originated from the upper domain that satisfy the Helmholtz equation takes the form of,

$$p_{0+} = a_+ e^{-ik_1 x_1} e^{-ik_{2,0}(x_2-h)} + a_- e^{-ik_1 x_1} e^{ik_{2,0}(x_2-h)}. \tag{1.1}$$

The pressure field in the porous layer is expressed as a sum of forward and backward waves,

$$p_p = b_+ e^{-ik_1 x_1} e^{-ik_{2,eq} x_2} + b_- e^{-ik_1 x_1} e^{ik_2(x_2-h)}, \tag{1.2}$$

and that in the lower air domain is,

$$p_{0-} = c e^{-ik_1 x_1} e^{ik_{2,0} x_2}, \tag{1.3}$$

with wavenumbers such that they hold for the Snell-Descartes law that states $k_1^2 + k_{2,0}^2 = k_0^2$ and $k_1^2 + k_{2,eq}^2 = k_{eq}^2$. The wavenumber should be imposed with $k_1 = k_0 \sin\theta$, $k_2 = -k_0 \cos\theta$, with θ the angle of incidence, that governs the direction of propagation of the incident field. The displacement fields are additionally expressed using the momentum equation relating the latter with the pressure fields as $\partial_2 p = -\mathrm{i}\omega \rho_i u_2, ::i =$.

Next, the interface conditions describe the relationships between properties at the boundary between two media, including the continuity or vanishing of given fields. If a media is terminated by a surface (e.g. a rigid wall), a constraint should be applied on the fields to represent it, denoted as boundary conditions. Assuming a continuity between the two fluids in the present case with interfaces located at $x_2 = 0$ and $x_2 = h$, we obtain the following conditions,

$$p^t(h) = p^l(h), \quad u_2^t(h) = u_2^l(h) \quad p^l(0) = p^b(0) \quad {}^l_2(0) = u_2^b(0), \tag{1.4}$$

with superscripts b and t denoting respectively the bottom and top interfaces.

The full analytical description of a system constituted from one or several interfaces can be done by writing a linear system built from the interface conditions, which vector of unknowns are the amplitudes of each of the existing waves in the system, that is $\begin{pmatrix} b_+ & b_- & a_+ & c \end{pmatrix}^\mathsf{T}$. The forcing right-hand side corresponds to the incident wave terms. The solution of the system yields all unknown amplitudes of the system, in particular the reflection and transmission coefficients. Imposing the interface conditions leads to writing the following matrix system,

$$\begin{pmatrix} 1 & e^{\mathrm{i}k_{2,eq}h} & 0 & -1 \\ \dfrac{k_{2,eq}}{\rho_{eq}} & -\dfrac{k_{2,eq}}{\rho_{eq}}e^{\mathrm{i}k_{2,eq}h} & 0 & \frac{k_{2,0}}{\rho_0} \\ e^{\mathrm{i}k_{2,eq}h} & 1 & -1 & 0 \\ \frac{k_{2,eq}}{\rho_{eq}}e^{\mathrm{i}k_{2,eq}h} & -\frac{k_{2,eq}}{\rho_{eq}} & -\frac{k_{2,0}}{\rho_0} & 0 \end{pmatrix} \begin{pmatrix} b_+ \\ b_- \\ a_+ \\ c \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ a_- \\ \frac{k_{2,eq}}{\rho_{eq}}a_- \end{pmatrix}. \tag{1.5}$$

vérifier l'écriture du système.

From this expression, the system can simply be solved to obtain all scattered amplitudes b_+, b_-, a_+ and c. In the case of a rigidly-backed layer, the interface at $x_2 = h$ is replaced by a rigid termination. For (equivalent) fluids, a hard wall or Neumann-type boundary condition is translated by a vanishing normal particle displacement field along the boundary, that is $u_2(\Gamma) = 0$. The term depending on c thus vanishes, since p_{0-} does not exist; this reduces the size of the system. In this case, the reflection coefficient R corresponds to the amplitude a_+.

This method is denoted in this work as the Global Method (GM), since all of the available data is used in the formulation of the problem as the unknowns are all wave amplitudes. As described in Section?? of Chapter ??, a general formalism can be used for this method and any stack of any layer can be described with the GM, by putting together a matrix system that contains all interface conditions in the system, in the general form,

$$\boldsymbol{M}(\omega)\boldsymbol{x} = \boldsymbol{f}, \tag{1.6}$$

where $\boldsymbol{M}$ corresponds to the interface conditions involved in the system, $\boldsymbol{x}$ contain the wave amplitudes and $\boldsymbol{f}$ is the forcing term, the incident plane wave amplitudes.

A few results for the simple cases of the scattering of a plane wave by a rigidly-backed layer and in the case of a symmetric configuration (with air coupling at both interfaces) are represented by the diagrams in Figs 1.1a and b and results in Figs 1.1c and d, respectively. The GM results are presented by the continuous lines. There, the layer for both porous and PEM are both $h = 5$ cm thick and the incident wave propagates with a $\theta = 30^o$ angle. The parameters for each material corresponds to the one used in Table ?? from Chapter ??. The system used to obtain the reflection coefficient for the PEM layer is done in a similar fashion as the equivalent fluid and can be read in the appendices of[2].

The first case, also referenced to as a reflection problem compares the equivalent fluid and the full Biot model response. The two curves follow the same trend but additional effects due to the skeleton motion can be observed on the continuous lines while the equivalent fluid response shows to be smoother. For all configurations, the melamine layer is acoustically transparent at lower frequency and absorption can be noticed at slightly higher frequencies.

1.1.2. Transfer Matrix Method

The widely used method we know today as the Transfer Matrix Method (TMM), was initially developed by the works of Thomson[3] and Haskell[4] in order to describe elastic wave propagation in multilayer systems. This method can be applied to a range of wave physics among geophysics, quantum mechanics, electromagnetics and obviously acoustics. TMM allows for an easy multiphysics coupling with interface matrices; its core lies in the evaluation of the state vector that regroups the variables of interest in the system and evaluate them at each interface, which is possible thanks to the interface conditions. Then the physical field amplitudes stored in the state vector are propagated from one layer to the next. The transfer matrix $\boldsymbol{T}$ relating the pressure and norrmal ddisplacement fields from the bottom (located at $x_2 = 0$) to the top interface (located at $x_2 = h$) of a fluid layer is thus,[5]

$$\begin{pmatrix} p(0) \\ u_2(0) \end{pmatrix} = \begin{pmatrix} \cos(k_{eq}h) & \mathrm{i}Z_{eq}\sin(k_{eq}h) \\ \mathrm{i}Z_{eq}^{-1}\sin(k_{eq}h) & \cos(k_{eq}h) \end{pmatrix} \begin{pmatrix} p(h) \\ u_2(h) \end{pmatrix}, \tag{1.7}$$

where k_{eq} is the bulk wavenumber of the wave propagating in the equivalent fluid and Z_{eq} is the medium impedances. The extension to PEM makes of a state vector that includes the fields, $\begin{pmatrix}\hat{\sigma}_{12} & u_2^s & u_2^t & \hat{\sigma}_{22} & p & u_1^s\end{pmatrix}^t ransp$. More details are given in Chapter 11 of Allard and Atalla textbook[6].

For stacks of layers, an amount of N layers of the same nature are easy to manipulate since $\boldsymbol{T}_{\text{global}} = \prod_{n=0}^{N} \boldsymbol{T}_n$ since they have the same size. multiphysics with interface matrix that relate the relation between the various fields at an interface (non square matrices). This method was deep dived in several textbooks among which we propose Refs.[6,7] for additional details. The TMM also allows for the computation of complex system by an assembly of elements including admittance matrices in order to consider Helmoltz Resonator (HR) embedded in an equivalent fluid layer[8].

While very versatile, it remains unstable in limit cases at high frequencies or for large thickness of the layers. While the formulation is mathematically exact, the evaluation of the system involves involves finite-precision calculation which leads the very sensitive exponential terms to diverge in the previously specified regimes. From the evaluation of the transfer matrix of the global system, the scattering matrix can be obtained and thus the reflection and transmission coefficients by simply manipulating the expressions for the fields contained in the state vector with the scattering coefficients to obtain that,

which is only valid for reciprocal and linear assumptions. The absorption coefficient for the same two configurations as the last part is presented in Figs 1.1c and d with the markers. It can be seen that the same results (up to the respectrive numerical precision of each method) are obtained.

1.1.3. Guided waves for dispersion problems

From the previously obtained linear systems in Section 1.1.1 for the configurations, and removing the incident wave from the system, i.e. the forcing (right-hand side) term, an homogeneous linear system is obtained, representing the guiding of waves in the layer. This set of equations becomes singular when its determinant vanishes for a given parameter, allowing to identify the modes of the structure. Several formalism can be adopted: solving for frequencies while imposing the wavenumber or the other way around. The first one is useful when interested in the time-dependant effects the structure has on the propagation. In this work, we are interested in the losses generated by the structure and the material and as such frequency is fixed in order to solve wavenumbers.

Computing the modes with semi-analytical tools

for porous/poroelastic materials, root-finding[9] + groby2008 single homogeneous layers

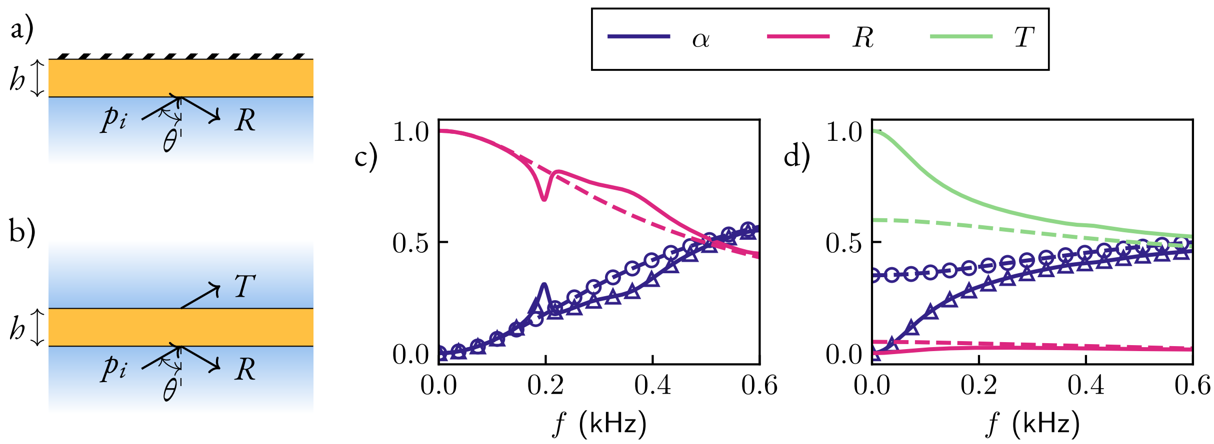

Fig. 1.1: Diagram for the a) reflection, b) transmission problem. Scattering coefficients from a poroelastic layer (continuous) and equivalent fluid layer (dashed) computed using the GM for the c) reflection, d) transmission problem. Results computed with the TMM are also shown for the absorption coefficient for the equivalent fluid (○) and for the absorption coefficient (△)

Once the linear system is obtained, we are faced with either or two of these possibilities. Solving wavenumbers can be done through the writing and solving of a (possibly generalized) eigenvalue problem or by root-finding of solutions in the case where writing the first one appears impossible.

Using Eq. (1.6), root finding such that $\det\left(\boldsymbol{M}(k_1, \omega)\right) = 0$; the determinant of the matrix from Eq. 1.5 can be expressed as

$$2\frac{k_{2,eq}k_{2,0}}{\rho_{eq}\rho_0}\cos(k_{2,eq}h) + \mathrm{i}\left(\frac{k_{2,eq}^2}{\rho_{eq}^2} + \frac{k_{2,0}^2}{\rho_0^2}\right)\sin(k_{2,eq}h) = 0 \tag{1.8}$$

Numerics for mode computations

Regarding the computation of complex modes in homogeneous layer, several numerical methods have been investigated in the last decades. Guided waves represents a large topic and we focus here on the case of simple geometries; similar configurations are considered, an unbounded single (or stack of) layers with two infinite interfaces as presented in the section where we were interested in their acoustic response.

Spectral methods have been around for quite some time and have been used for many numerical applications. We focus for this work on Spectral Collocation Method (SCM), which has been especially used for the computation of modes for guided waves in layers.[10]

In the case of simple 2D geometries, a 1D discretization along the transverse direction associated to a harmonic motion in the longitudinal direction. Even if the use of finite elements seems farfetched, this method could be relevant in contexts where boundary effects or guide with varying cross-sections are studied. However, a variation of finite elements based on a 1D discretization of the waveguide is entitled Semi-analytical Finite Elements (SAFE) and is akin to SCM which we have described in the previous paragraph. The 1D discretization is done along the thickness of the layer by regular nodes, and the fields are interpolated using finite elements shape functions. Examples of Semi-analytical Finite Elements (SAFE) use cases are very similar to SCM and include lister des publis ici. Examples are also given in Chapter ??. A detailed description of the calculation of Lamb modes using a simple SAFE implementation using high-order Lobatto polynomials is proposed in Appendix. ??. There, its performance is also compared to that of SCM.

discuss surface and guided waves in a paragraph

Fig. 1.2: Discretization for each method

A bestiary of the wave solutions in dissipative and open waveguides

In conservative and closed systems, propagating waves are either guided or trapped waves, while in open non-dissipative systems, the acoustic energy is able to radiate outside of the waveguide. Part of propagating waves are thus attenuated and radiate energy to the open domain. They are thus considered quasi-guided since they are propagating waves with some attenuation. Trapped modes can still occur. New family of modes appear as leaky backward and leaky forward waves that exhibit unique properties. Their counterparts are incoming waves.

The addition of dissipation leads to the mixing of several loss mechanisms: the leakage of waves and the dissipations within the guide (which can be of whatever nature). In this case, the separation between incoming and leaky waves is not possible anymore. This particular aspect has been very scarcely tackled in the literature: an unpublished paper is interested in an elastic waveguide[11] while in optics, a paper discusses the waves guided in nanospheres with a complex permittivity[12]vérifier la citation.

There is a discussion to be made on the relevance of the imposition of a local coupling contrary to a full modelling of the surrounding media with some termination condition satisfying the Sommerfeld condition, PML or analogous.

while leading to no assumptions on the radiating field nature contrary to imposing a local coupling, the modeling of the half-space by using some artificial absorbing layer at the end leads to the apparition of modes of the cavities[13]

1.1.4. Numerics for the computation of dispersion relations

- presentation of scm, safe + possible with fem
- cite chap. 2

1.2 Methods for modelling metaporous surfaces and interfaces

1.2.1. A word on periodic structures

A large literature that shows the enhancement of absorption properties by periodicizing porous layers with scatterers was discussed in the introduction. In this section, a presentation of the fundamental notions surrounding periodic structures is proposed. In wave physics, periodic structures have been shown to introduce interesting effects towards applications for wave control and constitute the cornerstone of acoustic metamaterials. The periodicity can be imposed in acoustic systems using the so-called Bloch-Floquet theorem. This formalism has been pioneered by Felix Bloch and Léon Brillouin among others[**bloch1929**, 14]. Throughout this manuscript, we consider a periodic cell along only the longitudinal direction, that 1D periodicity which we denote d. Taking an infinitely repeating lattice or a crystal, the description of the wave propagation in only one of the so-called unit cell, one spatial period of the total structure, is enough to describe the whole structure. The layer of infinite extent is thus limited to a single unit-cell. A wavefunction $\psi(\boldsymbol{x})$ describing the propagation should thus depend on a field with the same spatial periodicity as the lattice. To do so, the wavefunction is projected onto an infinite modal basis, that is Bloch modes,

$$\psi_q(\boldsymbol{x}) = e^{ikx_1}. \tag{1.9}$$

The wavefunction at the left boundary is thus related to that at the right boundary modulated by the spatial periodicity and some translation vector, that is $\psi(x + nd) = \psi(x)e^{i\boldsymbol{k}}$.

The possibility of reduction of the spatial considerations of the problem also leads to a reduction of the reciprocal wavenumber space. In periodic structure, we often restrict this reciprocal space to the first Brillouin zone, that is to wavenumbers restricted to $k_1 \in [0, \pi/d]$ when looking at the band structure. The consequences of periodic media are directly observable in the band structure: bands are usually folded when the wavenumber reaches this limit of the reciprocal space. This folding occurs at the Bragg frequency corresponds to a relation between the wavelength λ of the excitation signal and the periodicity of the crystal lattices, $d\sin(\theta) = n\lambda$, with θ the angle of incidence. This condition corresponds to destructive interferences: this phenomenon is a cause of the appearance of the famous bandgaps; in these bands of frequency, waves are unable to propagate. more on bg and stuff**stuff here**.

1.2.2. Scattering of waves by periodic structures: overview of methods

Analytical methods for the evaluation of the wave propagation in periodic structures often require tedious derivations, as the frequency regimes of interest is usually high enough to require the description of the multiple scattering happening in the structure. The scatterer of the unit cell onto which the incident wave impinges acts as a secondary source, which scattered field is successively scattered by each neighboring obstacles and so on. Multiple Scattering Theory is a theoretical framework that has been developed to account for these phenomena and was used to compute the acoustic response of metaporous structures with rigid inclusions[15] and metaporoelastic surfaces with solid and shell inclusions[2]. Both configurations have been tuned to show a perfect absorption peak in the frequency spectrum.

The remaining papers discussing how incident pressure fields are affected and scattered by metaporous structures use FEM, for the most part through the interface of Comsol Multiphysics. This very versatile and powerful method has been used for decades in all fields of physics and thus won't be detailed in this thesis. It is recommended to the reader as an introduction on the basis of FEM, the textbooks from Atalla and Sgard[16] related to acoustics and vibration problems and that of Dhatt *et al.* for higher-order finite elements problems[17]. This method allows for the discretization of the geometry via a mesh in order to solve the appropriate motion equations. This has allowed to use rather complex geometries, e.g spirals[18] and labyrinthine paths[19] as scatterers embedded in porous layers as discussed in the state of the art. Additionally, Gaborit *et al.*[20] have developed a method in order to couple FEM with Bloch modes expansion of analytical fields described by TMM. They have used it to compute the scattering coefficients of structure that include a periodic cell sandwiched by homogeneous layers with an embedded inclusion.

1.3 Dispersion relations and band structure in periodic lattices

One of the most popular methods for the computation of band structures in periodic systems made of classical lossless materials are the Plane Wave Expansion (PWE) which is semi-analytical. It relies on a Fourier transform of the motion equations, which allows an expression of the fields expanded onto Bloch modes, truncated for numerical applications. This method, while easy to implement presents important limitations due to the convergence of Fourier series and the matching of constituent materials for the lattice: materials should preferably of the same nature for the calculation to go well[7]. Generalized as the Extended Plane Wave Expansion (EPWE), the method also allows to obtain the evanescent Bloch modes in Ref.[21], allowing for complex dispersion diagram investigation and to account for evanescent modes in finite 2D periodic lattices.

The complexity of the system prohibits the use of such semi-analytical methods. For more complex geometries, we must turn to alternative methods. As the 1D discretization used by the SCM and SAFE and the plane wave formalism used in TMM and the GM is not capable of providing a description of the geometry of inhomogeneous layers, in our case, the effects of the scatterer in the unit cell on wave propagation.

1.3.1. Unpromising adaptation of existing methods

The complexity of the system of interest in this thesis as presented in Section?? of the Introduction prohibits the use of such semi-analytical methods. For more complex geometries, we must turn to alternative methods: as the 1D discretization used by the SCM and SAFE and the plane wave formalism used in TMM and the GM is not capable of providing a description of the geometry of inhomogeneous layers, in our case, the effects of the scatterer in the unit cell on wave propagation.

Multiple scattering could also be considered. However, it has been successfully used to describe the acoustic scattering of periodically arranged rigid inclusions in equivalent fluid unit cells[15] as well as solid inclusions in a poroelastic unit cell[2]. However, the so-called Schlömilch series that appear in the calculation, from the effect of the scattering of the neighboring cylindrical inclusions appear to be not so manipulable when considering complex longitudinal wavenumbers. Its expression is

$$\sigma_q^X = \sum_{l=1}^{\infty} H_q^{(1)}(k^X ld)\left((-1)^q e^{ik_1^X ld} + e^{-ik_1^X ld}\right), \tag{1.10}$$

and indeed, separating the k_1^X real and imaginary inevitably leads to at least one growing exponential term in the series, leading to it being unable to converge, up to our study.

The WFE as well as FEM has also been widely applied to these geometries. Their modularity and versatility lead them to be good candidates for the modelling of our structure.

We face challenges when trying to model this structure using the tools classically used in the literature, that is COMSOL Multiphysics. Although there are built-in tools in the software for ”*Eigenfrequency*” studies, the library aims at solving complex frequencies of the system, given a set of parameters for the wavenumber. As it has been discussed earlier in the chapter, the goal of this work is to solve for complex wavenumbers and real frequencies. Even trying a manual implementation of the model in the ”*Weak Form PDE*” interface, the tendency of this software to function as a black box introduces a new degree of complexity to this strategy. While FEM remains the soundest strategy for the modeling of the complex inhomogeneous geometry of the unit cell of metaporous surface, we propose to adapt in-house existing frameworks in order to develop numerical tools in this thesis.

trucs à rajouter jsp où

summary of all thesis of aberkane (soft solids), magliacano (disp w fem of closed system with porous layers), + gaborit et voir pour les autres

à distribuer au cours du chapitre

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